Subnetting Matrix Guide

This is a guide to using the Subnetting Matrix chart, one of the easiest ways (for most people) to work through subnetting problems.

There are two main phases to subnetting using the Subnetting Matrix. The last part of this document will include a walkthrough of using the Subnetting Matrix table to solve a subnetting problem

The Discovery phase is where we gather and organize information. It can be compared to an architect drawing the plans for a new house

The Execution phase is where we do the actual work of mapping out the networks created from subnetting. This is the phase where the carpenters, plumbers and electricians do the actual work of building the house based on the information from the Discovery phase.

Discovery Phase

1. Determine the number of bits to borrow
2. Define the interesting octet
3. Calculate the new subnet mask and CIDR
4. Determine the Block Size (aka Network Size)

Execution Phase

1. Map out the networks.
2. Determine the Broadcast Address (BC) for each network
3. Determine the range of Usable Addresses for each network

Axioms :

1. There are ONLY two types of bits in the Subnet Mask - Host Bits and Network Bits.
2. The Subnet Mask only gives you one bit of information - what part of its associated IP address represents the Network address and what part represents Host addresses
3. A value of 255 in an octet of the Subnet Mask means the corresponding octet in the IPv4 address represents the network
4. A value of 0 in an octet of the Subnet Mask means that the corresponding octet in the IPv4 address represents host addresses
5. A Subnet Mask octet with any value other than 0 or 255 means that the octet has BOTH Network and Host bits. This is called the Interesting Octet.
6. The first and last IPv4 addresses in a range can not be used for hosts or interfaces. The first address will always be the network number, the last address will always be the Broadcast Address

Definitions:

Interesting Octet - The ONLY octet in a Subnet Mask that

- Has a value that isn't 255 or 0
- Contains both Host and Network bits
- Is where all the "action" happens

Block Size - the size of each subnet created by subnetting. It can also be referred to as the Network Size or Subnet Size

Subnetting - the process of "borrowing" host bits in the Subnet Mask and converting them to Network Bits. This is done to conserve IPv4 addresses and divide one network into multiple networks

Example

Every subnetting problem will start with a defined requirement. Let's work through a problem using everything we have learned about using the Subnetting Matrix chart . Open the Subnetting Matrix chart and refer to it while working through this problem

Problem - For the following IP address, borrow enough bits to create at least 6 subnets

200.200.200.0/24 255.255.255.0

Solution:

Discovery Phase

1. Determine the number of bits to borrow

Look under column 2 of the Subnetting Matrix .
If we borrow 1 bit, we get 2 subnets - This doesn't meet the requirements
If we borrow 2 bits, we get 4 subnets , which doesn't meet our requirement
If we borrow 3 bits, we get 8 subnets , which meets our requirements of at least 6 subnets

Answer = 3 bits borrowed

2. Define the interesting octet

Our original network is 200.200.200.0 255.255.255.0. Look at the subnet mask “255.255.255.0”. Octet 1, 2 and 3 have a value of 255, which means all of those bits are network bits. The value of 0 in octet 4 shows that it is all host bits . We can only borrow host bits.
This means we will borrow bits from octet 4, and octet 4 will then have both Host and Network bits.

Answer - Interesting octet will be octet 4

3. Calculate the new subnet mask and CIDR

We borrowed three bits, so we look at column 3 of the Subnet Matrix to find a value of 224. This value will be inserted into the Interesting Octet of the Subnet Mask, to give us the following result :

200.200.200.0 255.255.255.224

We started with a CIDR of /24. We borrowed 3 bits , which gives us a CIDR of /27

4. Determine the Block Size (aka Network Size)

Knowing that we borrowed 3 bits, consult the Subnetting Matrix chart and see that our block size is 32.

This leaves us with the starting network of 200.200.200.0 255.255.255.224, with block size of 32

Execution Phase

1.Map out the networks.
This part is simple
Add the Block Size to the value in the Interesting Octet. Repeat
Remember that in this problem that the 4th octet is our Interesting Octet

1st network 200.200.200.0

Add 32 to 0 to get :
2nd network 200.200.200.32

Add 32 to 32 to get:
3rd network 200.200.200.64

Add 32 to 64 to get
4th network 200.200.200.96

Add 32 to 96 to get :
5th network 200.200.200.128

Add 32 to 96 to get:
6th network 200.200.200.160

And one more , add 32 to 160 to get:
7th network 200.200.200.192

2. Determine the Broadcast Address (BC) for each network

Now, designate the last address in each range as the Broadcast Address (BC). It is the IPv4 address that comes right before the next network address

1st network 200.200.200.0

1st BC = 200.200.200.31
2nd network 200.200.200.32

2nd BC = 200.200.200.63
3rd network 200.200.200.64

3rd BC = 200.200.200.95
4th network 200.200.200.96

4th BC = 200.200.200.127
5th network 200.200.200.128

5th BC = 200.200.200.159
6th network 200.200.200.160

6th BC = 200.200.200.191

7th network 200.200.200.192

3. Determine the range of Usable Addresses for each network

1st network 200.200.200.0
1st usable 200.200.200.1
Last usable 200.200.200.30
1st BC = 200.200.200.31

2nd network 200.200.200.32
1st usable 200.200.200.33
Last usable 200.200.200.62
2nd BC = 200.200.200.63

3rd network 200.200.200.64
1st usable 200.200.200.65
Last usable 200.200.200.94
3rd BC = 200.200.200.95

4th network 200.200.200.96
1st usable 200.200.200.97
Last usable 200.200.200.126
4th BC = 200.200.200.127

5th network 200.200.200.128
1st usable 200.200.200.129
Last usable 200.200.200.158
5th BC = 200.200.200.159

6th network 200.200.200.160
1st usable 200.200.200.161
Last usable 200.200.200.190
6th BC = 200.200.200.191

7th network 200.200.200.192